



AD23431 - STATISTICAL ANALYSIS AND COMPUTING**EXP 9: R methods for Train and Test data split**

Select a dataset of your choice (either built-in datasets like iris, mtcars, or any real-world dataset).

Perform the following tasks:

1. Data Preparation

- Load the dataset into R
- Display the structure and summary of the dataset

Answer:

```
data(iris)
```

```
str(iris)
```

```
summary(iris)
```

Output:

```
'data.frame': 150 obs. of 5 variables:
```

```
$ Sepal.Length: num  5.1 4.9 4.7 ...
```

```
$ Sepal.Width : num  3.5 3 3.2 ...
```

```
$ Petal.Length: num  1.4 1.4 1.3 ...
```

```
$ Petal.Width : num  0.2 0.2 0.2 ...
```

```
$ Species : Factor w/ 3 levels "setosa","versicolor","virginica"
```

summary:

```
Sepal.Length Sepal.Width Petal.Length Petal.Width Species
```

```
Min. :4.3 Min. :2.0 Min. :1.0 Min. :0.1 setosa :50
```

```
1st Qu.:5.1 1st Qu.:2.8 1st Qu.:1.6 1st Qu.:0.3 versicolor:50
```

```
Median :5.8 Median :3.0 Median :4.4 Median :1.3 virginica :50
```

```
Mean :5.84 Mean :3.05 Mean :3.76 Mean :1.20
```



2. Train–Test Split using Three Methods

Split the dataset into 70% training data and 30% testing data using the following approaches:

a) Base R Method

Answer:

```
set.seed(123)
index <- sample(1:nrow(iris), 0.7*nrow(iris))
train1 <- iris[index, ]
test1 <- iris[-index, ]
dim(train1)
dim(test1)
table(train1$Species)
table(test1$Species)
```

Output:

```
[1] 105  5
[1] 45  5
setosa versicolor virginica
35    36    34
setosa versicolor virginica
15    14    16
```



b) Using dplyr Package

Answer:

```
library(dplyr)
set.seed(123)
train2 <- sample_frac(iris, 0.7)
test2 <- anti_join(iris, train2)
dim(train2)
dim(test2)
table(train2$Species)
table(test2$Species)
```

Output:

```
[1] 105  5
[1] 45   5
```

c) Using caTools Package

```
library(caTools)
set.seed(123)
split <- sample.split(iris$Species, SplitRatio = 0.7)
train3 <- subset(iris, split == TRUE)
test3 <- subset(iris, split == FALSE)
dim(train3)
dim(test3)
table(train3$Species)
table(test3$Species)
```

Output:

```
[1] 105  5
[1] 45   5
setosa versicolor virginica
35    35    35
setosa versicolor virginica
15    15    15
```



3. Verification

- Display the dimensions of training and testing datasets for all three methods
- Compare the distribution of the target variable in train and test sets

Answer:

```
dim(train1)
dim(test1)
table(train1$Species)
table(test1$Species)
```

```
dim(train2)
dim(test2)
table(train2$Species)
table(test2$Species)
```

```
dim(train3)
dim(test3)
table(train3$Species)
table(test3$Species)
```

Output:

```
[1] 105  5
[1]  45  5
```

```
setosa versicolor virginica
  35      35      35
```

```
setosa versicolor virginica
  15      15      15
```